# Cross-Family and Partial-Family Prediction in WPM

**Normalization, compatibility, latent coordinates and selective transfer**

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**Abstract**

This study formulates transfer between heterogeneous source families as a sequence of coordinate, target-semantic and forecasting-eligibility checks. It distinguishes alternative forecasts of the same target from additive forecasts of different target components, derives a component-wise router with abstention, and gives sufficient conditions for risk transport. A known-family reference can reveal a shared latent coordinate only under calibration, branch and sampling assumptions. Worked examples show valid partial transfer, nonlinear target decoding, loss transport and structural aliasing. The treatment is mathematical and instructional; it does not assume universal cross-family invariance.

**Research lineage and mathematical status.** This study develops a WPM mathematical extension with reference to the author's published WPM architecture [1] and foundational n/s scaling-spectrum framework [2]. The n/s paper concerns conditional moment-spectrum parameterization; the WPM paper provides the broader architectural reference. The results proved here are mathematical consequences of stated assumptions, usually applications of standard linear algebra, probability or optimization. The WPM contribution is their organization into an explicit workflow. Modeling constructions and applicability conditions are identified separately; no universal forecasting superiority is asserted.

**Predictor inputs.** The forecasting layer described here is data driven: it learns from observed histories and matured outcomes. It is not fed the system's true governing equation or unobserved future states at issuance. Physical equations used below define observables, explain a limiting example or specify a separately labeled comparator. Optional supplied-map inverse problems and physics-informed variants must be declared separately; they are not evidence that the default forecast knows the generator.

## 1 What is a family, and what is transferred?

The reader should know regression, conditional distributions and the chain rule. A family is a declared set of systems or models sharing some mechanism or representation. It may be indexed by parameters $\theta_f$, a state law $\dot{x} = f_f(x, u; \theta_f)$, or a conditional distribution. "Same family" is therefore relative to a modeling description, not a guarantee of identical forecasting performance.

For source family $f$, component $g$ and lens $k = (W, m)$, denote a candidate by $h_{fgk}$. Its proposed contribution must have a well-defined mathematical role. We distinguish:

1. *Alternative estimators*: several candidates estimate the same scalar, vector or probability target; averaging may be meaningful.
2. *Additive components*: separate predictions estimate different terms in $Y = \sum_g Y_g$; terms must be assembled, not averaged away.
3. *Different targets*: endpoint state, cumulative event, scale and a hidden component are different objects even when they are measured from one run.

Compatibility must be checked at this level. Removing an unsupported component from a required sum changes the target unless its contribution is known to vanish or a valid fallback supplies it.

Let $\mathcal{F}_t$ denote all information available at issuance. The requested physical duration $q$ is fixed. If sampling interval is $\Delta_f$, a sample-count horizon is $P_f = q/\Delta_f$ when this is meaningful; noninteger values require interpolation or a continuous-time predictor. A coordinate or clock estimate must not silently change $q$.

## 2 Two normalizations with different duties

Define system normalization $N_{S,f} : X_f \to Z$ and target normalization $N_{T,f} : Y_f \to U$. The first aligns observed state or system descriptors; the second aligns what a prediction means. A family-specific event threshold may require a physical rescaling even after features have been standardized.

**Worked example 2.1** (Time and target scale)**.** Suppose $x_f$ has natural amplitude $a_f > 0$ and relaxation time $\tau_f > 0$. Use $z = x_f/a_f$ and $u = t/\tau_f$. The same dimensionless forecast duration $h$ corresponds to physical durations $q_f = h\tau_f$, not equal physical durations across families. If the user requests a fixed physical $q$, each family must evaluate its corresponding $h_f = q/\tau_f$. An event $x_f > b_f$ becomes $z > b_f/a_f$; equal numerical values of $b_f$ need not represent equal events.

Normalizers may be physically supplied or estimated causally. Estimated $a_f, \tau_f$ carry uncertainty. A target map that is not injective can destroy information: $N_T(y) = y^2$ identifies $y$ and $-y$. No inverse normalization can recover that lost sign without additional restrictions.

**Proposition 2.2** (Means and nonlinear decoding)**.** *If $U = N_T(Y)$ and $g = N_T^{-1}$ exists, the physical conditional mean is $\mathbb{E}[g(U) \mid \mathcal{F}_t]$. It is generally not $g(\mathbb{E}[U \mid \mathcal{F}_t])$.*

*Proof.* Conditional expectation commutes with affine functions but not with general nonlinear functions. For $U$ taking 0 and $\log 4$ with equal probability and $g(u) = e^u$, $\mathbb{E}g(U) = 5/2$, while $g(\mathbb{E}U) = 2$. □

For squared physical-space error, use the first quantity. For quantiles under a strictly increasing transformation, transform the quantiles directly. For event probabilities, transform the event set. For a full distribution, push forward the law. These are distinct decoding rules.

## 3 A compatibility contract before forecasting

For each candidate define three issuance-time masks:

$$I_{fgk}(t, T, q) = I_{fg}^{\text{phys}}\, I_{fgk}^{\text{target}}(T, q)\, I_{fgk}^{\text{qual}}(t, T, q). \tag{3.1}$$

$I^{\text{phys}}$ checks that the component exists and has a meaningful interpretation. $I^{\text{target}}$ checks target, units, endpoint and loss semantics after normalization. $I^{\text{qual}}$ checks that the modeling assumptions and available qualification support using this candidate at the requested horizon and context. These are logical decisions or defined policies, not automatic probabilities of correctness.

Physical absence, unobserved presence and unqualified forecasting are different states. An absent mode is not a noisy zero-frequency observation. A present mode with unknown future forcing is not a qualified predictor merely because it can be measured now.

**Application rule.** Perform the contract before merging landscapes or weighting predictions. Preserve compatible individual pockets and independent regimes. When qualification is unavailable, abstain or invoke an independently appropriate fallback rather than assigning zero risk.

For a scalar target with all candidates estimating the same normalized $U$, choose nonnegative base weights $w_{fgk}$. If $Z_t = \sum I_{fgk} w_{fgk} > 0$, the routed mixture is

$$\widehat{U} = \sum_{fgk} \frac{I_{fgk} w_{fgk}}{Z_t} \widehat{U}_{fgk}. \tag{3.2}$$

If $Z_t = 0$, the expression is undefined. Returning zero is an invented forecast, not abstention.

## 4 The correct rule for partial-family components

Suppose the physical target decomposes as $Y = \sum_{g=1}^{G} Y_g$, with all components required. Build an eligible alternative set $\mathcal{E}_g$ separately for every $g$. Then

$$\widehat{Y}_g = \sum_{(f,k)\in\mathcal{E}_g} \pi_{fgk} \widehat{Y}_{fgk}, \qquad \sum_{(f,k)\in\mathcal{E}_g} \pi_{fgk} = 1, \quad \widehat{Y} = \sum_g \widehat{Y}_g. \tag{4.1}$$

The units of $Y_g$ and the assembly law must be known. For a nonlinear observation $Y = H(Y_1, \ldots, Y_G)$, apply $H$ to a joint predictive law or justified point estimates, propagating dependence and nonlinear uncertainty. Merely summing normalized components may not decode the original target.

**Worked example 4.1** (Partial compatibility without changing the target)**.** Let $Y = R + Z$. Source A supplies qualified forecasts $R_A = 2$ but has no model for $Z$. Source B supplies $R_B = 4$ and $Z_B = 1$. With equal eligible weights within the $R$ component, $\widehat{R} = 3$; the only eligible $Z$ forecast is 1, so $\widehat{Y} = 4$. Averaging the three numbers $2, 4, 1$ gives $7/3$ and confuses component assembly with alternative estimation. If B's $Z$ forecast is unqualified and $Z$ is required, total-target prediction must abstain or supply a separate fallback for $Z$.

**Proposition 4.2** (Componentwise squared-error control)**.** *For component errors $e_g = \widehat{Y}_g - Y_g$, $(\sum_g e_g)^2 \le G \sum_g e_g^2$. More generally total MSE equals $\sum_{g,h} \mathbb{E}[e_g e_h]$.*

*Proof.* Cauchy–Schwarz applied to $\langle (1, \ldots, 1), e \rangle$ yields the first inequality. Expanding the square and taking expectation gives the second identity. Small component errors can reinforce or cancel through their cross moments. □

Thus component qualification is necessary to justify the assembly, but error dependence still matters for total uncertainty.

# 5 A theorem about when risk really transports

Let $(Z, U)$ denote normalized features and target. Use source and destination laws $P_s, P_d$. The following result is a sufficient condition, not a test that can be certified from a small sample.

**Theorem 5.1** (Conditional-law transport)**.** *Assume $P_d(U \mid Z) = P_s(U \mid Z)$, destination support is covered by source support, and a loss $\ell(U, h(Z))$ is integrable. If $w(Z) = dP_d^Z/dP_s^Z$, then*

$$R_d(h) = \mathbb{E}_d \ell(U, h(Z)) = \mathbb{E}_s[w(Z)\ell(U, h(Z))]. \tag{5.1}$$

*Proof.* Write $R_d = \int\int \ell(u, h(z)) P_d(du \mid z) P_d^Z(dz)$. Replace the conditional law by $P_s(du \mid z)$ and the marginal by $w(z)P_s^Z(dz)$. The resulting iterated integral is the source weighted expectation. □

This is the covariate-shift identity underlying importance weighting [5]. It fails when the conditional target law changes, when destination support has no source counterpart, or when the normalizer changes the event meaning. Large density ratios create high variance even if the identity is valid. Estimated weights add estimation error; clipping them trades bias for variance.

**Worked example 5.2** (Changing mixture proportions)**.** Let $Z \in \{0, 1\}$, source probabilities $(1/2, 1/2)$ and destination probabilities $(1/4, 3/4)$. Suppose the candidate's conditional risks are $(1, 3)$ under the shared conditional law. Source risk is 2 and destination risk is $5/2$. Weights $(1/2, 3/2)$ give $\frac{1}{2} \cdot \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{3}{2} \cdot 3 = 5/2$. Equal normalized feature names alone would not justify equal weighting.

# 6 Cross-family recovery through a known coordinate

A different kind of transfer supplies additional measurements. Suppose

$$A(t) = a(q(t)), \qquad B(t) = b(q(t); \theta), \tag{6.1}$$

where the calibrated reference $a$ is known and $q$ is a shared latent coordinate. If $a$ is injective over the relevant domain, $q(t) = a^{-1}(A(t))$ and one can fit $b$ in the recovered coordinate. This can distinguish an apparent nonstationarity in observation time from a simpler law in $q$.

**Proposition 6.1** (Local coordinate conditioning)**.** *If a scalar reference has $|a'(q)| \geq c > 0$ on a monotone interval, then inverse-coordinate perturbations satisfy $|\delta q| \leq |\delta A|/c$ for observations staying in that interval. For a vector reference with derivative $J_a(q)$, local first-order error is $\delta q = J_a(q)^+ \delta A + o(\|\delta A\|)$ when the derivative has full column rank.*

*Proof.* The scalar mean value theorem gives $|a(q_1) - a(q_2)| \geq c|q_1 - q_2|$. For the vector statement, linearize $\delta A = J_a \delta q + o(\|\delta q\|)$ and apply the left pseudoinverse. Global branches are not resolved by a local derivative condition. □

If the reference is itself noisy, $q$ is a noisy regressor. Ordinary least squares treating it as exact can be biased. A joint observation model for A and B, or an appropriate errors-in-variables method, is required for inferential uncertainty.

# 7 Phase branches and sampling aliases

A common oriented reference is

$$A(q) = (\cos \omega q, \sin \omega q), \qquad \phi(t) = \operatorname{atan2}(A_2, A_1). \tag{7.1}$$

The observed phase determines $\omega q$ only modulo $2\pi$. Consecutive increments are uniquely recoverable in a specified principal interval if the true phase increments are known to satisfy $|\omega(q_{j+1} - q_j)| < \pi$. An initial phase branch remains a separate anchor. Noise near the branch cut or an increment near $\pi$ requires uncertainty over multiple wraps.

**Worked example 7.1** (A known coordinate and a different equation)**.** Let $q(t) = t + t^2/10$, reference frequency $\omega = 4$, and $B(t) = 2 + 3q(t) + q(t)^2/2$. After branch-resolved phase inversion, the three observations at $q = 0, 1, 2$ give $B = (2, 11/2, 10)$. Fitting columns $(1, q, q^2)$ recovers $(2, 3, 1/2)$: the second difference equals 1, so the quadratic coefficient is $1/2$, the first increment gives the slope, and $B(0)$ fixes the level.

Predicting $B$ at a future physical endpoint also requires $q(t + P)$. Recovering the conditional law $b(q)$ does not reveal a future unobserved coordinate. If the earlier equation for $q(t)$ was not supplied, it must be estimated using the issuance prefix.

# 8 Relational discovery, excitation and independent readings

For a coupled model write

$$\dot{A} = f_A(A, u) + \sum_j \theta_j \phi_j(A, B), \tag{8.1}$$

$$\dot{B} = f_B(B, u) + \sum_j \gamma_j \psi_j(A, B). \tag{8.2}$$

Fit compatible conditions jointly with shared relational coefficients and condition-specific initial states when appropriate. Integral residuals can avoid direct differentiation of noisy signals. A prior for shared coefficients enters once in the joint model, not once for each trajectory as if the coefficient were redrawn.

Measurement precision and distinguishability are different. Independent readings of the same state reduce measurement variance. Diverse inputs can change the sensitivity design. For $Y_r = X + \eta + \epsilon_r$, with independent zero-mean $\epsilon_r$ of variance $\sigma^2$ and common error $\eta$ of variance $\tau^2$,

$$\operatorname{Var}(\bar{Y} - X) = \tau^2 + \sigma^2/M. \tag{8.3}$$

The common-error floor does not vanish. For independent conditions with sensitivity matrices $J_e$ and noise covariances $\Sigma_e$, local information is $\mathcal{I} = \sum_e J_e^\top \Sigma_e^{-1} J_e$. Replication scales information in existing directions; excitation can create missing directions.

**Worked example 8.1** (Breaking an exact alias)**.** If $B(t) = e^{ct}$, then $A\dot{B} = cAB$. A dictionary with columns $AB$ and $A\dot{B}$ is rank deficient on this trajectory, regardless of sensor precision. Two normalized design rows $(1, -0.7)$ and $(1, 0.3)$ have determinant 1 and separate two coefficients. Repeating only the first row preserves rank one.

## 9 Mode choice and uncertainty after routing

Fixed selects a declared eligible lens. Multi combines several eligible alternatives. Auto updates weights from causal context and matured losses. The compatibility mask is not a fourth mode and renormalizing after a mask is not itself evidence of adaptive learning.

For compatible distribution forecasts $F_i$ of the same target, $F = \sum_i \pi_i F_i$ is a valid mixture CDF when $\pi_i \geq 0$ and $\sum_i \pi_i = 1$. Its mean is $\mu = \sum_i \pi_i \mu_i$, and

$$\mathrm{Var}(Y) = \sum_i \pi_i \{\sigma_i^2 + (\mu_i - \mu)^2\}. \tag{9.1}$$

This follows by conditioning on the mixture label and expanding $\mathbb{E}(Y^2) - \mu^2$. It is not the variance of an average of simultaneously realized source components; that uses their joint covariance. Calibrated source distributions can also lose calibration after routing based on changing contexts. Proper scoring and conditional checks remain necessary [4].

## 10 A transfer worksheet with physical decoding

Consider two source families measuring a common dimensionless state Z in different physical units: $X_A = 2Z$, $X_B = 5Z$. Their targets are future physical states in the same respective units. Suppose source predictions of the common future Z are 1 and 3, with weights 3/4 and 1/4. Normalize first: $\widehat{Z} = 3/2$. For a destination with physical amplitude 4, the decoded point forecast is 6. Directly averaging the unnormalized source numbers 2 and 15 would give 21/4; that number has no declared destination-target meaning.

Now suppose the normalized forecasts are distributions, not points. Let $Z_A$ have mean 1 and variance 1, and $Z_B$ mean 3 and variance 4. The mixture mean is 3/2, within-component variance is 7/4, and between-component variance is 3/4, giving total variance 5/2. Under the linear destination map $Y = 4Z$, mean is 6 and variance is $16(5/2) = 40$. If the destination map were nonlinear, propagate the full distribution rather than multiplying the mean and variance by a constant.

Timing must also align. Suppose source A has $\Delta_A = 1$ second per sample and source B has $\Delta_B = 1/2$ second. A physical query $q = 3$ seconds corresponds to horizons 3 and 6 samples. Using P=3 in both would average forecasts for different endpoints. If one source is qualified only for two seconds, it does not participate in the three-second query merely because a three-sample forecast is available.

Finally, distinguish missing target components from missing alternatives. If A and B each estimate the whole Z, an ineligible B can simply be excluded and A's weight renormalized. If A estimates one term and B alone estimates a second required term, excluding B leaves an incomplete total-target model. The same numerical mask has different semantic consequences in these two situations.

## 11 A quantitative local compatibility bound

Exact equality of normalized conditional laws is strong. One useful weaker statement compares a destination target U with a transported source target $U_s$ under a specified coupling. Assume $|U - U_s| \leq \delta_T$ almost surely, normalized features satisfy $\|Z - Z_s\| \leq \delta_S$, and h is K-Lipschitz. If the scalar loss is L-Lipschitz in both its target and prediction arguments in the sense

$$|\ell(u, a) - \ell(v, b)| \leq L(|u - v| + |a - b|), \tag{11.1}$$

then

$$|R_d(h) - R_s(h)| \le L(\delta_T + K\delta_S) \tag{11.2}$$

for the coupled distributions under consideration. The proof applies the pointwise inequalities and then takes expectations. This is a sufficient perturbation bound under an explicit coupling, not an empirical compatibility detector. Squared loss is not globally Lipschitz on an unbounded domain; bounded targets and forecasts or a different moment argument are needed.

If source risk is known only within $\varepsilon_s$, a permissible destination-risk budget B can be certified by $\widehat{R}_s + \varepsilon_s + L(\delta_T + K\delta_S) \le B$ when every bound is valid. The certificate fails if one simply guesses small transport errors from visual similarity of pocket shapes. Stating the residual uncertainty is part of the method.

An explicit bridge also clarifies error propagation. If $B = b(q;\theta)$, local perturbations are

$$\delta B \simeq \partial_q b\, \delta q + \nabla_\theta b^\top \delta\theta. \tag{11.3}$$

Variance includes the covariance between $\delta q$ and $\delta\theta$ when the same data estimate both. A precisely fitted conditional equation can have poor operational forecasts when the future coordinate is uncertain. Conversely, a robust direct predictor may forecast B well without uniquely identifying $b(q;\theta)$.

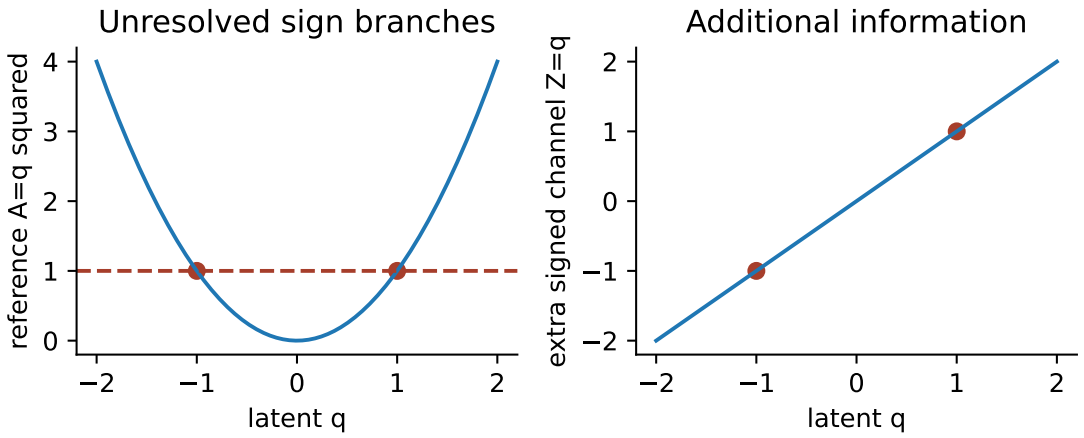



Figure 1: A squared reference is globally two-to-one. A separate signed channel distinguishes the displayed branches. Additional information, rather than a different label for the same squared observation, removes the ambiguity.

## 12 What partial compatibility does not permit

Partial-family routing is not an instruction to manufacture a common physical family. It preserves differences that remain meaningful. If two systems use different coordinate gauges, event definitions or observation mechanisms, a common feature vector may still fail to support a shared target law. Standardization removes units; it does not establish invariance.

Qualification at several discrete horizons also does not prove qualification at every intermediate duration. An interpolation guarantee needs regularity in horizon, a bound on interpolation error and a preserved target definition. Similarly, a reference phase can be locally invertible and globally aliased; confidence in local fitting does not settle wrap count.

When returning a partial forecast, identify its coverage in target space: which components are predicted, which are absent by definition, which are unknown and which are withheld by the

qualification policy. This is more informative than returning a single "incompatible family" flag. Preserve separate pocket and regime records so that later observations can refine the appropriate component without erasing the others.

## 13 A general application recipe

1. Write the requested physical target and endpoint. State which terms are additive and which candidates are alternatives.
2. Define $N_S$ and $N_T$ separately, including units, invertibility and causal estimation. Transform events as well as numbers.
3. Preserve each family's useful landscape layers, individual pocket flow, position and existence. Do not collapse independent modes to force agreement.
4. Check physical validity, target semantics and qualification at each component–pocket level. Record rejection reasons.
5. Assemble required components using (4.1); average only compatible estimators of the same target. Abstain if a required component is uncovered.
6. Where a reference supplies a latent coordinate, check calibration, branches, sampling and sensitivity before inversion.
7. Fit relations using sufficient excitation and a declared dictionary. Distinguish search failure, numerical fitting failure and indistinguishable alternatives.
8. Propagate coordinate, coefficient, forcing and observation uncertainty to the physical target. Keep calibration separate from similarity of normalized landscapes.

Do not infer a universal common equation from similar pocket locations, force a missing component into a zero-valued mode, interpolate horizon eligibility without assumptions, or count a deterministic pocket summary as an extra sensor.

## 14 Exercises with solution trails

**Exercise 14.1.** Two sources estimate the same normalized target as 2 and 6 with base weights 1 and 3. The second is ineligible. Find the routed forecast and explain the empty-set case.

*Solution.* Masking gives denominator 1 and forecast 2. If both are excluded, the denominator is zero and the router abstains. Retaining the old denominator would produce $1/2$ and would not be a normalized mixture.

**Exercise 14.2.** A required target is $Y = Y_1 + Y_2$. Only $Y_1$ has an eligible predictor. Is reporting that predictor as $\widehat{Y}$ justified?

*Solution.* Only if a justified constraint says $Y_2 = 0$, or another qualified model supplies its contribution. Otherwise this is a partial-target forecast and must be labeled as such.

**Exercise 14.3.** For $A(q) = q^2$, determine where a coordinate inverse is unique and well-conditioned.

*Solution.* Globally, $q$ and $-q$ are aliased. On $q \geq a > 0$, the positive branch is unique and $|a'(q)| \geq 2a$, so inverse error is bounded by measurement error divided by $2a$. Conditioning deteriorates as $a \to 0$.

## Scope of the mathematical exposition

Unless explicitly identified as a sourced evidence statement, numerical values are constructed arithmetic examples. Application requires specifying the observation process, fitting unknown quantities, and checking reliability for the intended population and horizon. The archival DOI for this study is 10.5281/zenodo.23782422.

## References

[1] T. Nguyen. *The WPM Framework: A Generalized Mathematical Architecture for Cross-Scale, Spatial, and Dynamical Prediction*. Zenodo, 2026. doi:10.5281/zenodo.23248983.

[2] T. Nguyen. *An n/s Scaling-Spectrum Framework for Generalized Moment Scaling: Curvature, Signed-Moment Asymmetry, and Prospective Moment-Order Validation*. Zenodo, 2026. doi:10.5281/zenodo.23111783.

[3] S. Boyd and L. Vandenberghe. *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares*. Cambridge University Press, 2018. https://stanford.edu/~boyd/vmls/.

[4] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* 102(477), 359–378, 2007. doi:10.1198/016214506000001437.

[5] M. Sugiyama, M. Krauledat and K.-R. Müller. Covariate shift adaptation by importance weighted cross validation. *Journal of Machine Learning Research* 8, 985–1005, 2007. https://jmlr.org/papers/v8/sugiyama07a.html.